

Bitcoin: Consensus, Relay Policy, and Node Configuration

A deep dive into node behavior: validation, relay choices, and configuration.



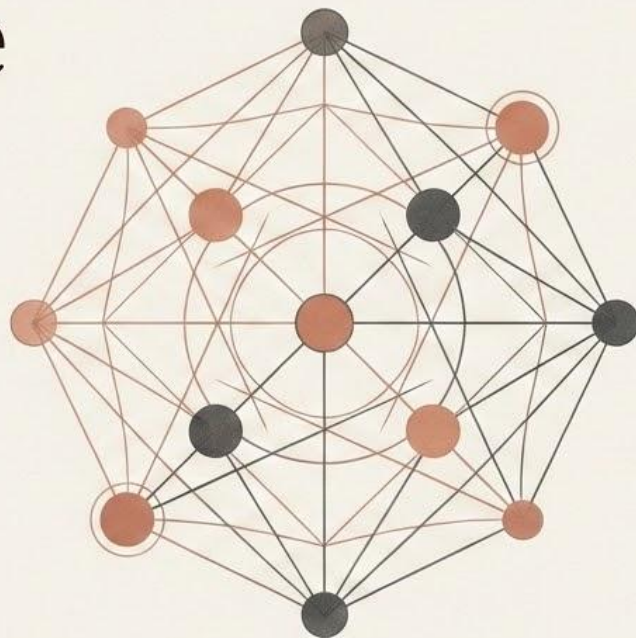
Consensus rules — what Bitcoin must enforce



Relay policy — what nodes choose to relay



bitcoin.conf — how we control node behavior

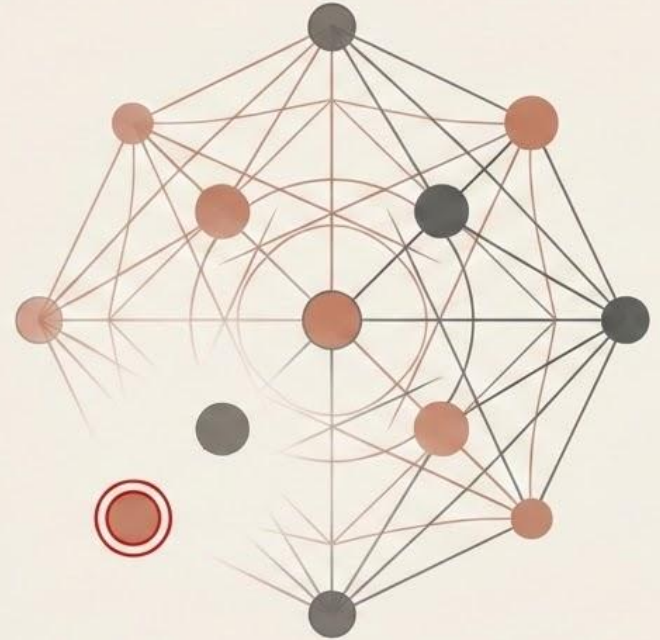


What is Consensus?

 The set of rules every node agrees on to determine what is valid.

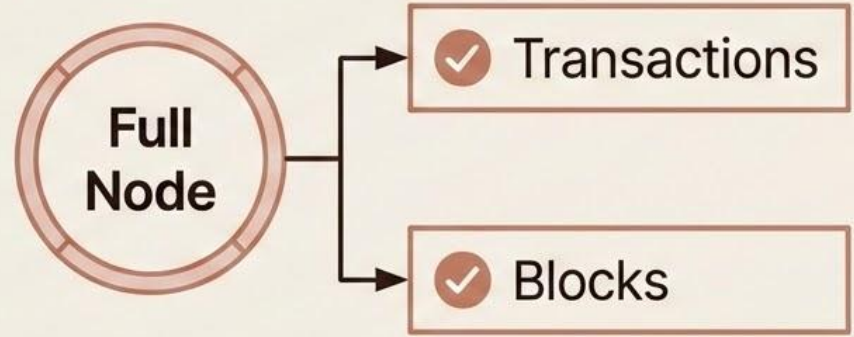
 These rules are not optional.

 If your node breaks consensus rules, it is no longer part of the Bitcoin network.



Key Idea

Consensus = validation
rules for blocks and
transactions



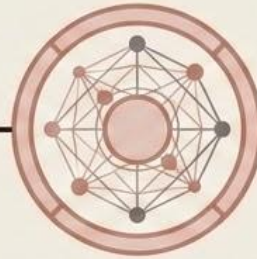
⚙️ **No trust.**

⚙️ **No central authority.**

What does consensus check?

✓ Transactions

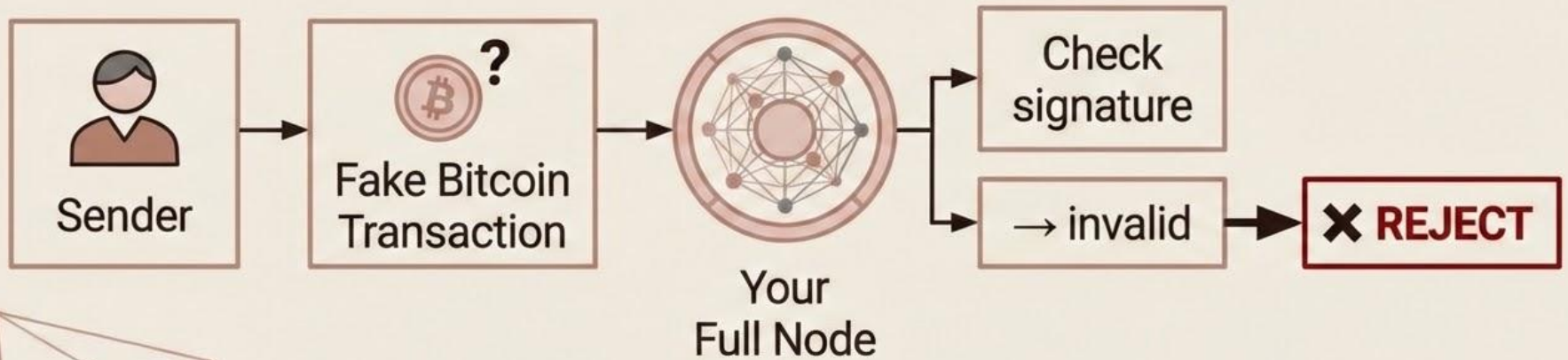
- ✿ Signatures are valid
- ✿ Inputs are not already spent (no double spend)
- ✿ Scripts evaluate to true
- ✿ Amounts are correct



📦 Blocks

- ✿ Block hash meets difficulty target
- ✿ Block size/weight is valid
- ✿ All transactions inside are valid
- ✿ Block connects to previous valid block

Example (make it intuitive)



That transaction never enters your reality.

Important property

Consensus is:

- ✿ Deterministic - Same input yields the same output every time.
- ✿ Strict - Requires exact agreement among all participants.
- ✿ Global - Consensus spans the entire network, not just a part.

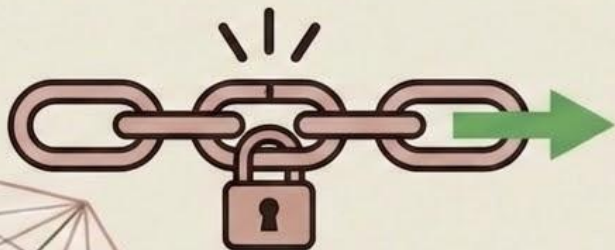


Every node must reach the same result.

Hard Fork vs Soft Fork

Soft Fork:

- ⚙ Tightens rules
- ⚙ Backward compatible



Hard Fork:

- ⚙ Loosens rules
- ⚙ Not backward compatible

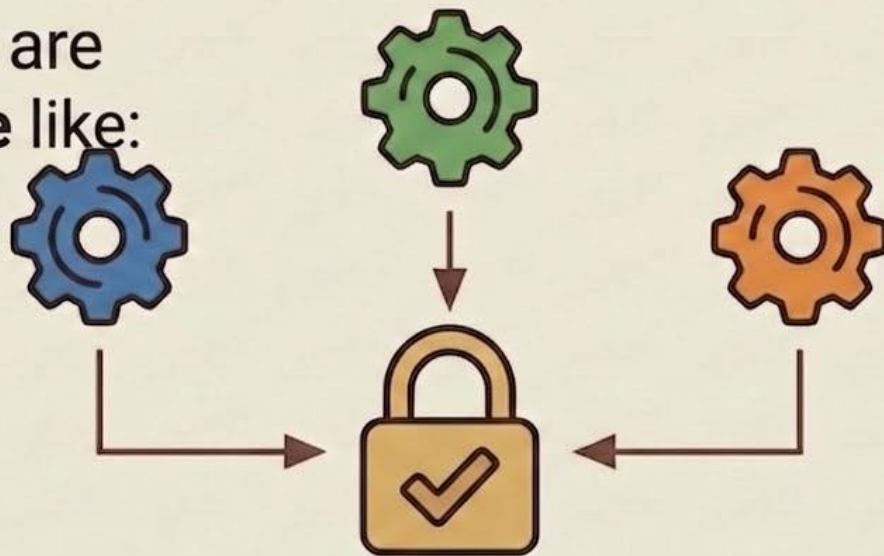


Bitcoin strongly prefers soft forks.

Real implementations

These consensus rules are implemented in **software** like:

- ⚙ Bitcoin Core
- ⚙ btcd
- ⚙ bcoin



Even though **implementations differ**, they must produce the same **validation results**.

Examples of Bitcoin Core Consensus Rules

Key rules from the `src/consensus` directory that define the Bitcoin network protocols.



Block Validation (`consensus.h` & `merkle.h`)

- **MAX_BLOCK_WEIGHT:**
Limits block size to prevent spam.
- **Merkle Root Verification:**
Ensures transaction integrity within a block.
- **Block Subsidy Halving Interval:** Determines when mining rewards decrease.



Transaction Verification (`tx_verify.h`)

- **Prevents Double-Spending:**
Ensures an input is not used more than once.
- **Signature Verification:**
Confirms transaction authorization.
- **Dust Limit:** Discourages uneconomical small outputs.

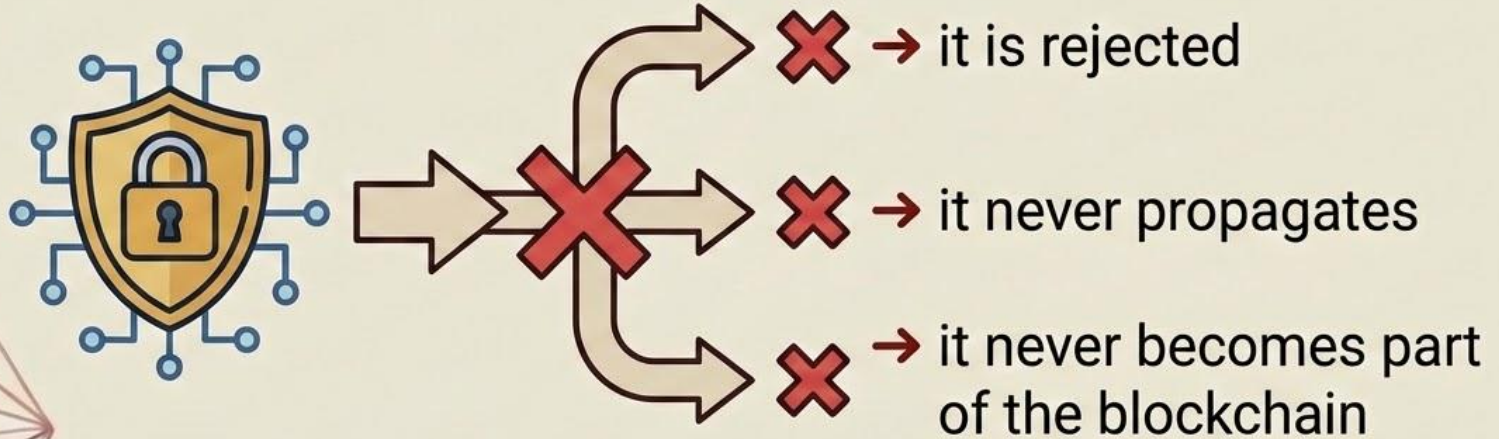


Network Parameters (`params.h`)

- **Difficulty Adjustment Algorithm:** Maintains block discovery time.
- **Proof-of-Work Limit:** Sets the minimum difficulty.
- **Genesis Block Definition:**
The starting point of the blockchain.

Key Takeaway: Consensus Defines Truth in Bitcoin

Consensus defines truth in Bitcoin.



The slide features a light beige background with decorative geometric patterns in the corners. These patterns consist of interconnected lines and dots in shades of grey, orange, and blue, forming a network-like structure. At the top, there are three horizontal bars in blue, gold, and dark red. A thin dark red horizontal line is positioned below the main text.

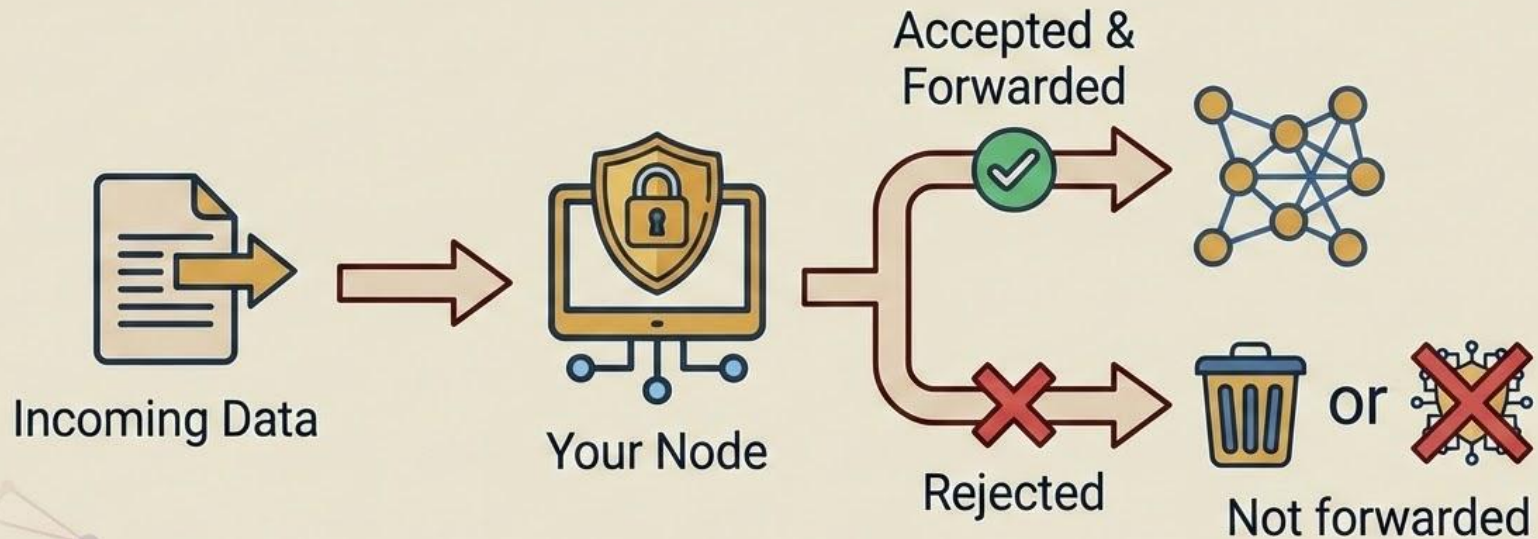
**Now we move to
something different**



Part 2: Relay Policy

What is Relay Policy?

Relay policy = what your node chooses to forward to others.



Key Difference

Consensus	Policy
 Must follow	 Can choose

Concept Meaning

Why policy exists

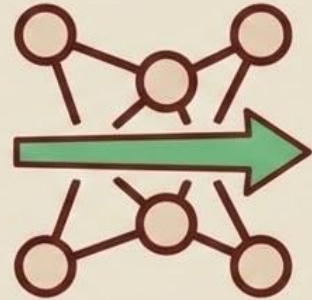
Why not just relay everything valid?



Spam protection



DoS resistance



Network efficiency

Examples of relay policy



Minimum fee (minrelaytxfee)

Transactions below
fee threshold
→ not relayed



Standardness rules

Only “standard”
transaction types
are relayed



Mempool limits

Node limits
memory usage



Replace-by-Fee (RBF)

Allows replacing
transactions with
higher fee

Important insight



**Valid
(consensus)**

Follows network's
core rules.

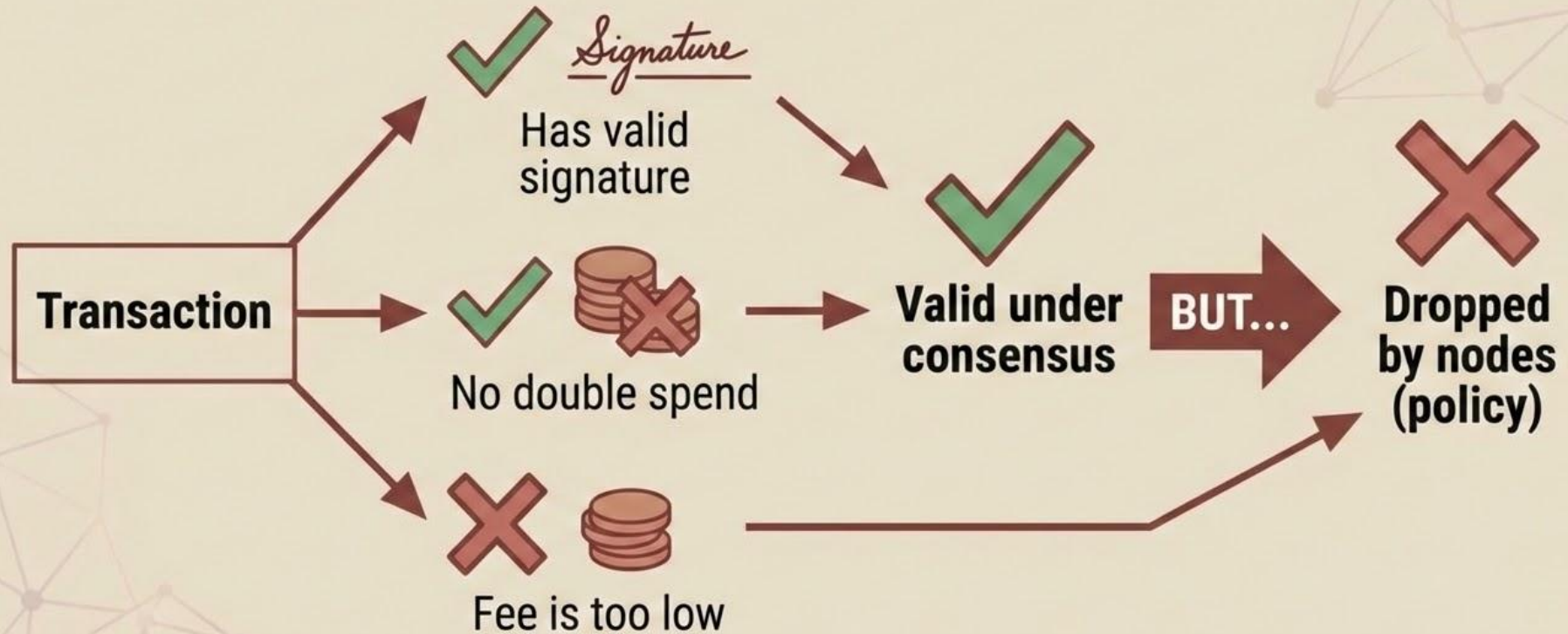


**Not relayed
(policy)**

Rejected by a node's
local rules.

**This is one of the most
misunderstood concepts.**

Real example



Why this matters & Node independence

Policy shapes:



• Mempool behavior



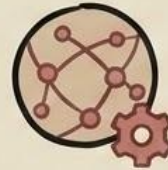
• Fee market



• Transaction propagation



Node independence



Node A
(Policy X)



Node B
(Policy Y)

Each node can have different policies.



But consensus must match globally.

Key takeaway

Policy is:



Local



Flexible



Protective

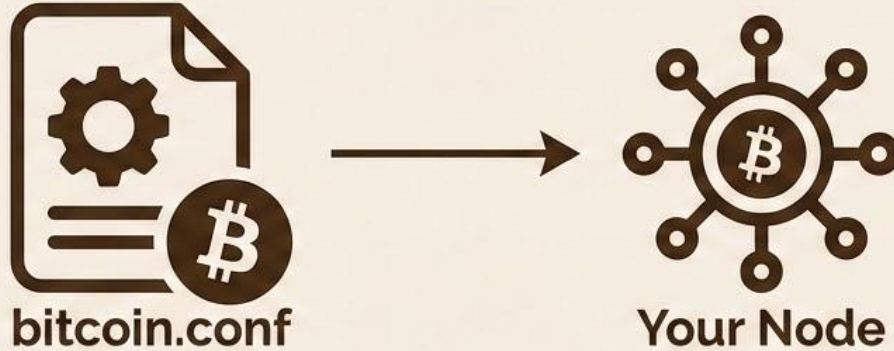


It helps keep the network efficient
without breaking consensus.



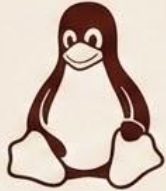
Part 3: Bitcoin.conf

What is bitcoin.conf?



bitcoin.conf is the configuration file used to control your node. It allows customization of node behavior and policy, defining parameters such as network connection, resource usage, and consensus rules.

Where it lives



Linux: `~/.bitcoin/bitcoin.conf`



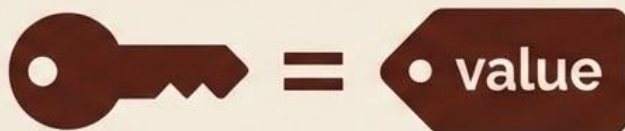
macOS: `~/Library/Application
Support/Bitcoin/bitcoin.conf`



Windows:
`%APPDATA%\Bitcoin\bitcoin.conf`

Basic structure

It's just key-value pairs:



`option=value`

Core settings

- **Network:**
`testnet=1`
`regtest=1`
- **RPC:**
`server=1`
`rpcuser=youruser`
`rpcpassword=yourpassword`
- **Connections:**
`maxconnections=40`

Relay policy settings



Minimum relay fee:

`minrelaytxfee=0.00001`



Mempool size:

`maxmempool=300`

Performance tuning

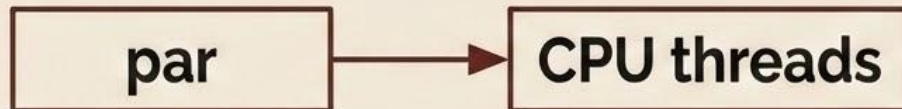
dbcache

dbcache=1000



par

par=4



Privacy & network

proxy

proxy=127.0.0.1:9050

onlynet

onlynet=onion

- This routes traffic through Tor.



Practical Example Config



```
server=1  
txindex=1  
daemon=1  
maxconnections=50  
dbcache=1000  
minrelaytxfee=0.00001
```

Important warning



- Misconfiguring your node can:
 - Reduce performance
 - Break connectivity
 - Affect privacy

Real-world usage

- Developers use config for:
 - Running APIs
 - Lightning nodes
 - Custom mempool policies
 - Custom mempool policies



Key takeaway

- Control behavior
- Tune performance
- Customize policy



Final summary



Consensus

- Global rules
- Defines validity



Relay Policy

- Local decisions
- Controls propagation



bitcoin.conf

- Node customization
- Controls behavior

If consensus defines truth, and policy defines behavior,
then configuration defines your node's personality.

Q&A

Thank you — I'm happy to take questions.